1. Chapter 8, problem 4(a),(b),(c),(d).
2. Chapter 8, problem 16(a),(b).
3. Chapter 8, problem 53.

Chapter 8, problem 4(a),(b),(c),(d)

(a). First, note

$$E[X] = \frac{2}{3}\theta \cdot 0 + \frac{1}{3}\theta \cdot 1 + \frac{2}{3}(1-\theta) \cdot 2 + \frac{1}{3}(1-\theta) \cdot 3 = \frac{7}{3} - 2\theta.$$

Then we compute the sample mean

$$\bar{X} = \frac{1}{10}(3+0+2+1+3+2+1+0+2+1) = \frac{3}{2}.$$

Hence,

$$\widehat{\theta}_{mom} = \frac{1}{2}\left(\frac{7}{3} - \bar{X}\right) = \frac{1}{2}\left(\frac{7}{3} - \frac{3}{2}\right) = \frac{5}{12}.$$

(b). We first calculate the variance of X.

$$\begin{aligned}
\operatorname{Var}[X] =& E\left[X^2\right] - (E[X])^2 \\
=& \frac{2}{3} \cdot 0^2 + \frac{1}{3}\theta \cdot 1^2 + \frac{2}{3}(1-\theta)\, 2^2 + \frac{1}{3}(1-\theta)\, 3^2 - \left(\frac{7}{3} - 2\theta\right)^2 \\
=& \frac{2}{9} + 4\theta - 4\theta^2.
\end{aligned}$$

Hence

$$\operatorname{Var}\left[\hat{\theta}_{mom}\right] = \operatorname{Var}\left[\frac{1}{2}\left(\frac{7}{3} - \bar{X}\right)\right] = \operatorname{Var}\left[\frac{1}{2}\bar{X}\right] = \frac{1}{4n}\operatorname{Var}[X_1] = \frac{1}{4n}\left(\frac{2}{9} + 4\theta - 4\theta^2\right).$$

Next, plugging in $n = 10$ and $\hat{\theta}_{mom} = \frac{5}{12} \simeq 0.417$ yields an approximate standard error for $\hat{\theta}_{mom}$ of 0.173.

(c). The likelihood function is

$$\operatorname{lik}(\theta) = \left(\frac{2}{3}\theta\right)^2 \left(\frac{1}{3}\theta\right)^3 \left[\frac{2}{3}(1-\theta)\right]^3 \left[\frac{1}{3}(1-\theta)\right]^2 = \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right)^5 \theta^5 (1-\theta)^5.$$

Hence, the log-likelihood function is

$$\ell(\theta) = 5\log\frac{2}{3} + 5\log\frac{1}{3} + 5\log\theta + 5\log(1-\theta)$$

so that

$$\ell'(\theta) = \frac{5}{\theta} - \frac{5}{1-\theta} \Rightarrow \theta = \frac{1}{2}.$$

Since $\ell''(\theta) = -\frac{5}{\theta^2} - \frac{5}{(1-\theta)^2} < 0$, it follows that $\theta = \frac{1}{2}$ indeed maximizes the likelihood function. Thus, $\widehat{\theta}_{mle} = \frac{1}{2}$.

(d). Define

$$Y_k = \sum_{i=1}^{n} 1\{X_i = k\}$$

for $k = 0, 1, 2, 3$. Notice that

$$\text{lik}(\theta) = \left(\frac{2}{3}\right)^{Y_0+Y_2} \left(\frac{1}{3}\right)^{Y_1+Y_3} \theta^{Y_0+Y_1}(1-\theta)^{Y_2+Y_3}$$

Hence $\ell'(\theta) = 0$ is equivalent with

$$\frac{Y_2 + Y_3}{1-\theta} = \frac{Y_0 + Y_1}{\theta}$$

which is equivalent with $\theta = (Y_0 + Y_1)/n$ using $Y_0 + Y_1 + Y_2 + Y_3 = n$. Hence

$$\widehat{\theta}_{MLE} = \frac{Y_0 + Y_1}{n}$$

Now, notice that $Y_0 + Y_1$ has a Binomial distribution with parameters n and $p(0) + p(1) = \theta$ as we consider n trials and we consider observing a 0 or a 1 a success. Consequently,

$$E[\widehat{\theta}_{MLE}] = n\theta/n = \theta$$
$$\text{Var}(\widehat{\theta}_{MLE}) = \theta(1-\theta)/n$$

so that the standard error of $\widehat{\theta}_{MLE}$ is estimated by $\sqrt{1/40} \approx 0.158$.

Chapter 8, problem 16 (a) and (b)

(a) By symmetry around 0, we have

$$E[X] = \int_{\mathbb{R}} x f(x|\sigma)\, dx = 0.$$

Next,

$$E\left[X^2\right] = \int_{\mathbb{R}} x^2 \frac{1}{2\sigma} e^{-\frac{|x|}{\sigma}} dx = \int_0^{\infty} x^2 \frac{1}{\sigma} e^{-\frac{x}{\sigma}} dx = E\left[Y^2\right] = 2\sigma^2$$

for $Y \sim \text{Exponential}\left(\frac{1}{\sigma}\right)$ with

$$E[Y^2] = \text{Var}(Y) + (E[Y])^2 = 2\sigma^2.$$

Hence, it follows that

$$\sigma = \sqrt{E\left[X^2\right]/2}$$

and thus

$$\widehat{\sigma}_{mom} = \sqrt{\frac{1}{2n}\sum_{i=1}^{n} X_i^2}.$$

(b) We first write out the log-likelihood function

$$\begin{aligned}\ell(\sigma) &= \sum_{i=1}^{n} \left(-\log 2 - \log\sigma - \frac{|X_i|}{\sigma} \right) \\ &= -n\log 2 - n\log\sigma - \frac{\sum_{i=1}^{n} |X_i|}{\sigma}.\end{aligned}$$

so that

$$\ell'(\sigma) = -\frac{n}{\sigma} + \frac{\sum_{i=1}^{n} |X_i|}{\sigma^2} = 0 \Rightarrow \hat{\sigma}_{mle} = \frac{1}{n}\sum_{i=1}^{n} |X_i|.$$

Chapter 8, problem 53

(a). Since

$$E\left[X_1\right] = \int_{[0,\theta]} \theta^{-1} x dx = \frac{\theta}{2},$$

it follows that $\widehat{\theta}_{mom} = 2\bar{X}$. Also we have

$$\begin{aligned} E\left[X_1^2\right] &= \int_{[0,\theta]} \theta^{-1} x^2 = \left[\frac{x^3}{3\theta}\right]_0^\theta = \frac{\theta^2}{3} \\ \Rightarrow \text{Var}\left[X_1\right] &= E\left[X_1^2\right] - E\left[X_1\right]^2 = \frac{\theta^2}{3} - \frac{\theta^2}{4} = \frac{\theta^2}{12}, \end{aligned}$$

For the mean and variance of $\widehat{\theta}_{mom}$, note

$$\begin{aligned} E\left[\widehat{\theta}_{mom}\right] &= E\left[2\bar{X}\right] = \theta, \\ \text{Var}\left[\widehat{\theta}_{mom}\right] &= \text{Var}\left[2\bar{X}\right] = \frac{4}{n}\text{Var}\left[X_1\right] = \frac{\theta^2}{3n}. \end{aligned}$$

(b). We have

$$\text{lik}(\theta) = \frac{1}{\theta^n} \prod_{i=1}^n \mathbb{1}_{[0,\theta]}\left(X_i\right) = \frac{1}{\theta^n} 1\{X_i \leq \theta, \quad \text{for all } i\} = \frac{1}{\theta^n} 1\{X_{(n)} \leq \theta\}$$

where $X_{(1)} \leq \cdots \leq X_{(n)}$ denote the order statistics. Since θ^{-n} is a decreasing function of θ, $\text{lik}\left(\theta\right) = \theta^{-n} 1\{\theta \geq X_{(n)}\}$ is maximized for $\widehat{\theta}_{mle} = X_{(n)}$.

(c). We first get the density for $Y = X_{(n)}$. Note that, for $0 < y < \theta$,

$$\begin{aligned} F_Y(y) &= P\left(Y < y\right) \\ &= P\left(\max\left(X_1, \ldots, X_n\right) \leq y\right) \\ &= P\left(X_1 \leq y, \ldots, X_n \leq y\right) \\ &= \prod_{i=1}^n P\left(X_i \leq y\right) \\ &= \theta^{-n} y^n. \end{aligned}$$

It follows that Y has density

$$f_Y(y) = \frac{d}{dy} F_Y(y) = n\theta^{-n} y^{n-1} \qquad \text{for } 0 < x < \theta,$$

so that

$$E[\widehat{\theta}_{mle}] = E[Y] = \int_0^\theta y f_Y(y) dy = \int_0^\theta \theta^{-n} n y^n dy = \frac{n}{n+1}\theta \neq \theta.$$

Hence, $\widehat{\theta}_{mle}$ is a biased estimator of θ and the bias

$$b(\hat{\theta}_{mle}) = E[\hat{\theta}_{mle}] - \theta = -\frac{\theta}{n+1}$$

is asymptotically negligible.

Next, we compute the variance. Observe that

$$E[Y^2] = \int_0^\theta y^2 \frac{ny^{n-1}}{\theta^n} dy = \left[\frac{ny^{n+1}}{(n+2)\,\theta^n}\right]_0^\theta = \frac{n}{n+2}\theta^2.$$

It follows that

$$\begin{aligned}
\mathrm{Var}[\widehat{\theta}_{mle}] =& \mathrm{Var}(Y) = E[Y^2] - (E[Y])^2 = \frac{n\theta^2}{n+2} - \left(\frac{n\theta}{n+1}\right)^2 = \frac{n\theta^2}{(n+1)^2\,(n+2)};\\
\mathrm{MSE}\left[\widehat{\theta}_{mle}\right] =& \mathrm{Var}\left(\widehat{\theta}_{mle}\right) + b^2\left(\widehat{\theta}_{mle}\right) = \frac{n\theta^2}{(n+1)^2\,(n+2)} + \frac{\theta^2}{(n+1)^2} = \frac{2\theta^2}{(n+1)\,(n+2)};\\
\mathrm{MSE}\left[\widehat{\theta}_{mom}\right] =& \mathrm{Var}\left[\widehat{\theta}_{mom}\right] + b^2\left(\widehat{\theta}_{mom}\right) = \frac{\theta^2}{3n}.
\end{aligned}$$

Note that

$$\frac{\mathrm{MSE}\left[\widehat{\theta}_{mle}\right]}{\mathrm{MSE}\left[\widehat{\theta}_{mom}\right]} = \frac{6n}{(n+1)\,(n+2)} = \frac{6}{n+\frac{2}{n}+3} \overset{n\to\infty}{\longrightarrow} 0,$$

it follows that $\mathrm{MSE}\left[\widehat{\theta}_{mle}\right]$ decays much faster than $\mathrm{MSE}\left[\widehat{\theta}_{mom}\right]$.

(d). Let

$$\hat{\theta} = \frac{n+1}{n}\widehat{\theta}_{mle}.$$

Then

$$E[\hat{\theta}] = \frac{n+1}{n} E\left[\widehat{\theta}_{mle}\right] = \theta,$$

so $\hat{\theta}$ is an unbiased estimator of θ.